



The herbal extract ALS-L1023 from *Melissa officinalis* reduces weight gain, elevated glucose levels and β -cell loss in Otsuka Long-Evans Tokushima fatty rats

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ABSTRACT

Ethnopharmacological relevance: *Melissa officinalis* L. (Labiatae; lemon balm) is a traditional medicinal plant with hypoglycemic and hypolipidemic effects; however, how it imparts its beneficial effects remains unclear. We thus hypothesized that the herbal extract ALS-L1023, isolated from *Melissa officinalis*, inhibits obesity and diabetes, and tested our hypothesis using Otsuka Long-Evans Tokushima fatty (OLETF) rats, which are an established animal model of type 2 diabetes.

Materials and methods: In this study, 28-week-old OLETF rats were fed a high-fat diet for 4 weeks to induce a marked impairment of the insulin response and were treated with or without ALS-L1023. Subsequently, the variables and determinants of glucose metabolism and pancreatic function were assessed via blood analysis, histology, immunohistochemistry, and real-time polymerase chain reaction.

Results: The administration of ALS-L1023 resulted in a weight reduction without changes in food intake. It also markedly inhibited hyperglycemia and hypoinsulinemia, and restored β -cell mass that was severely impaired in OLETF rats. There was a decrease in lipid accumulation in the liver and skeletal muscle of the obese rats after treatment with ALS-L1023. Concomitantly, there was an increase in the expression levels of fatty acid-oxidizing enzymes (AMPK α 2, ACOX, MCAD, and VLCAD) in the liver and skeletal muscle after ALS-L1023 treatment. Furthermore, ALS-L1023 attenuated the pancreatic inflammation including the infiltration of CD68-positive macrophages and mast cells, in addition to attenuating the expression of inflammatory factors (IL-6 and CD68). **Conclusions:** These results suggest that treatment with ALS-L1023 may reduce weight gain, elevated glucose levels, and β -cell loss, by changing the expression of fatty acid-oxidizing enzymes in the liver and skeletal muscle, including inflammatory factors in the pancreas. These findings indicate that ALS-L1023 may be an effective therapeutic strategy to treat human obesity and type 2 diabetes.

1. Introduction

Obesity-associated metabolic disorders, such as insulin resistance and type 2 diabetes (T2D), are risk factors that threaten the health of people worldwide. Insulin resistance is a pre-diabetic disease and the most important marker of subsequent T2D (Sung et al., 2012). In the insulin-resistant state, the pancreatic islet β -cells compensate by secreting more insulin to maintain normoglycemia (Ahrén and Pacini, 2002). Over time, this compensatory β -cell response eventually fails.

Gradual reductions in β -cell function and glucose tolerance contribute to the development of T2D (Kahn, 2003; Prentki and Nolan, 2006)./

Insulin resistance and T2D are associated with ectopic lipid deposition. Excess ectopic lipid accumulation in the liver and skeletal muscle impairs insulin signaling, which leads to a reduced hepatic glycogen synthesis and to a decreased muscle uptake of glucose (Perry et al., 2014). The insulin resistance in the liver and muscle can be overcome by increasing insulin production in the pancreas, which results in hyperinsulinemia. However, pancreatic failure and reduced insulin production eventually occur during the progression to overt T2D (van Herpen

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Abbreviations

ACOX	acyl-CoA oxidase;
AMPK α 2	AMP-activated protein kinase α 2;
HFD	high-fat diet;
H&E	hematoxylin and eosin;
LETO	Long-Evans Tokushima Otsuka;
MCAD	medium-chain acyl-CoA dehydrogenase;
IL-6	interleukin 6;
OLETF	Otsuka Long-Evans Tokushima fatty;
PCR	polymerase chain reaction;
PPAR α	peroxisome proliferator-activated receptor α ;
T2D	type 2 diabetes;
TNF α	tumor necrosis factor α ;
VLCAD	very long-chain acyl-CoA dehydrogenase

and Schrauwen-Hinderling, 2008). Additionally, impaired fat oxidation is a predictor of both weight gain and T2D (Broskey et al., 2018). Dysregulation of fatty acid oxidation is a hallmark of T2D progression, and leads to an increase in lipid stores and insulin resistance (Krssak et al., 1999; McGarry, 2002).

Insulin resistance and β -cell dysfunction are induced by chronic inflammation associated with obesity in diabetic subjects. Obesity-associated inflammation is characterized by an increased abundance of macrophages and an enhanced expression of inflammatory cytokines in adipose tissues. The same process most likely occurs in the pancreas because it contains a pool of tissue-resident macrophages. Pancreatic islets in T2D models show evidence of inflammation, including an increase in immune cell infiltration and an elevation of inflammatory cytokine production (Donath et al., 2009; Marselli et al., 2013; Khodabandehloo et al., 2016). Increased islet-associated immune cells and islet-derived inflammatory factors were observed in human T2D patients, high-fat diet (HFD)-fed C57BL/6J mice, Goto-Kakizaki (GK) rats, and db/db mice (Ehses et al., 2007), suggesting that islet inflammation could be a key contributing factor in the development of T2D.

Melissa officinalis L. (Labiatae; lemon balm) is a traditional edible-medicinal herb used to treat mental and respiratory diseases, and used as a flavoring food additive for more than 2000 years (Blumenthal et al., 2000; Shakeri et al., 2016). According to the Iranian Traditional Medicine, *Melissa officinalis* is also crucial for treatment of diabetes and cardiovascular disease, which suggests it has potential hypolipidemic and hypoglycemic effects (Castellano et al., 2013; Weidner et al., 2014; Hasanein and Riahi, 2015; Shakeri et al., 2016); however, its underlying mechanisms of action remains unclear. Furthermore, our previous studies demonstrated that the herbal extract ALS-L1023, isolated from the leaves of *Melissa officinalis*, regulates obesity and nonalcoholic fatty liver disease in diet-induced obese C57BL/6J mice (Park et al., 2015; Woo et al., 2016; Kim et al., 2017), which suggests that ALS-L1023 may alleviate obesity and its related disorder T2D in obese diabetic Otsuka Long-Evans Tokushima fatty (OLETF) rats. Therefore, we examined the effects of ALS-L1023 on weight gain, blood glucose levels, and islet β -cell loss in HFD-fed OLETF rats. We further investigated the cellular and molecular mechanisms involved in these processes. Here, we report that ALS-L1023 may mitigate obesity and diabetes in obese diabetic OLETF rats, in part, through changes in the expression of fatty acid oxidative enzymes in the liver and skeletal muscle, and by attenuating inflammatory factors in the pancreas.

2. Materials and methods

2.1. Preparation of ALS-L1023

ALS-L1023 was prepared by a two-step organic solvent fractionation

from the leaves of *Melissa officinalis*, which were purchased from Alfred Galke GmbH (Harz, Germany). A voucher specimen for *Melissa officinalis* (AL-R001) was deposited at AngioLab, Inc. Briefly, the dried *Melissa* leaves were extracted with aqueous ethanol and the resulting extract was filtered and concentrated. The concentrated ethanol extract was further fractionated using ethyl acetate, and the resulting fraction was concentrated and dried to obtain ALS-L1023 in a dried powder form. The high-performance liquid chromatography profile of ALS-L1023 was previously described (Woo et al., 2016).

2.2. Animal studies

Twelve OLETF and four Long-Evans Tokushima Otsuka (LETO) rats at 28 weeks of age were randomly assigned to one of four groups: a non-obese LETO group fed a chow diet ($n = 4$; LETO), an obese OLETF control group fed an HFD ($n = 4$; OLETF-Con), and two groups of obese OLETF rats fed an HFD supplemented with 0.4% or 0.8% (w/w) of ALS-L1023 [$n = 4$; OLETF-ALS (0.4%) or $n = 4$; OLETF-ALS (0.8%), respectively] for a period of 4 weeks. Four or 8 g of the ALS-L1023 powder was mixed with 1 kg of the HFD. The body weight of each animal was measured three times a week by an individual blinded to the treatments. Food intake was determined by measuring the amount of food consumed by the rats throughout the treatment period. All the animal experiments were approved by the Institutional Animal Care and Use Committee of Mokwon University (permit number: NVRQS AEC-5) and followed National Research Council Guidelines.

2.3. Blood analysis

The serum concentrations of triglycerides and free fatty acids were measured using an automatic blood chemical analyzer (CIBA Corning, Oberlin, OH). Blood glucose concentrations were measured using the Accu-Chek Performa System (Roche, Basel, Switzerland). The serum insulin concentrations were analyzed using an enzyme-linked immunosorbent assay kit from Linco Research (St. Charles, MO, USA).

2.4. Histological analysis

For hematoxylin and eosin (H&E) staining, the tissues were fixed in 10% phosphate-buffered formalin for 24 h and processed to obtain paraffin-embedded blocks. Paraffin blocks were cut using a Leica microtome (Leica Microsystems GmbH, Wetzlar, Germany), and the tissue sections (5- μ m thick) were stained with H&E for light microscopic examination. To quantify the size of epididymal adipocytes, the H&E-stained sections were assessed using an Image-Pro Plus analysis system (Media Cybernetics, Bethesda, MD, USA). To examine the lipid accumulation in the liver and skeletal muscle, the liver and gastrocnemius muscle tissues were embedded in a frozen section compound (Leica Microsystems GmbH) and frozen in liquid nitrogen. Transverse sections (7 μ m) were cut using a cryostat microtome (Leica Microsystems GmbH) at -20°C and thawed on slides. Lipid accumulation in tissues was observed by staining with Oil red O for microscopic examination.

2.5. Immunohistochemistry

Immunohistochemistry was used to detect insulin and CD68. After epitope retrieval, pancreas sections (3- μ m thick) were incubated with monoclonal mouse anti-insulin (1:1400 dilution; I2018; Sigma-Aldrich, St Louis, MO, USA) and anti-CD68 (1:200 dilution; ab955; Abcam, Cambridge, UK) primary antibodies to detect insulin-secreting β -cells and infiltrated macrophages, respectively. The sections were then incubated with an anti-mouse IgG biotinylated secondary antibody (Vector Laboratories, Burlingame, CA, USA) using diaminobenzidine (Vector Laboratories) as the chromogen. Insulin-secreting β -cell areas were measured using ImageJ software (National Institutes of Health, Bethesda, MD, USA), and the relative insulin-positive areas were

expressed as percentages of the total pancreatic area occupied by β cells.

2.6. Quantitative real-time polymerase chain reaction

Total cellular RNA was prepared using TRI reagent (Molecular Research Center, Cincinnati, OH, USA). Complementary DNA (cDNA) was synthesized from total cellular RNA (2 μ g) using MG Moloney murine leukemia virus reverse transcriptase (CancerRop, Seoul, Korea). The genes of interest were amplified from the synthesized cDNA using AccuPower™ GreenStarMaster Mix (Bioneer, Daejeon, Korea) and an Exycycler™ 96 Real-Time Quantitative Thermal Block (Bioneer). The PCR primer pairs are presented in [Supplementary Table 1](#). The relative gene expression levels were calculated as the ratios of the target gene cDNA to β -actin cDNA.

2.7. Statistical analysis

All the values are expressed as means $\pm$ standard deviation. Statistical analyses were performed using analysis of variance with Turkey's post-hoc tests. A p value < 0.05 was considered statistically significant.

3. Results

3.1. ALS-L1023 inhibits visceral obesity without changing food consumption

OLETF rats fed an HFD supplemented with 0.8% ALS-L1023 had lower body weights after 4 weeks of treatment compared with the obese OLETF-Con rats ([Fig. 1A](#)). The addition of ALS-L1023, at 0.4% or 0.8%,

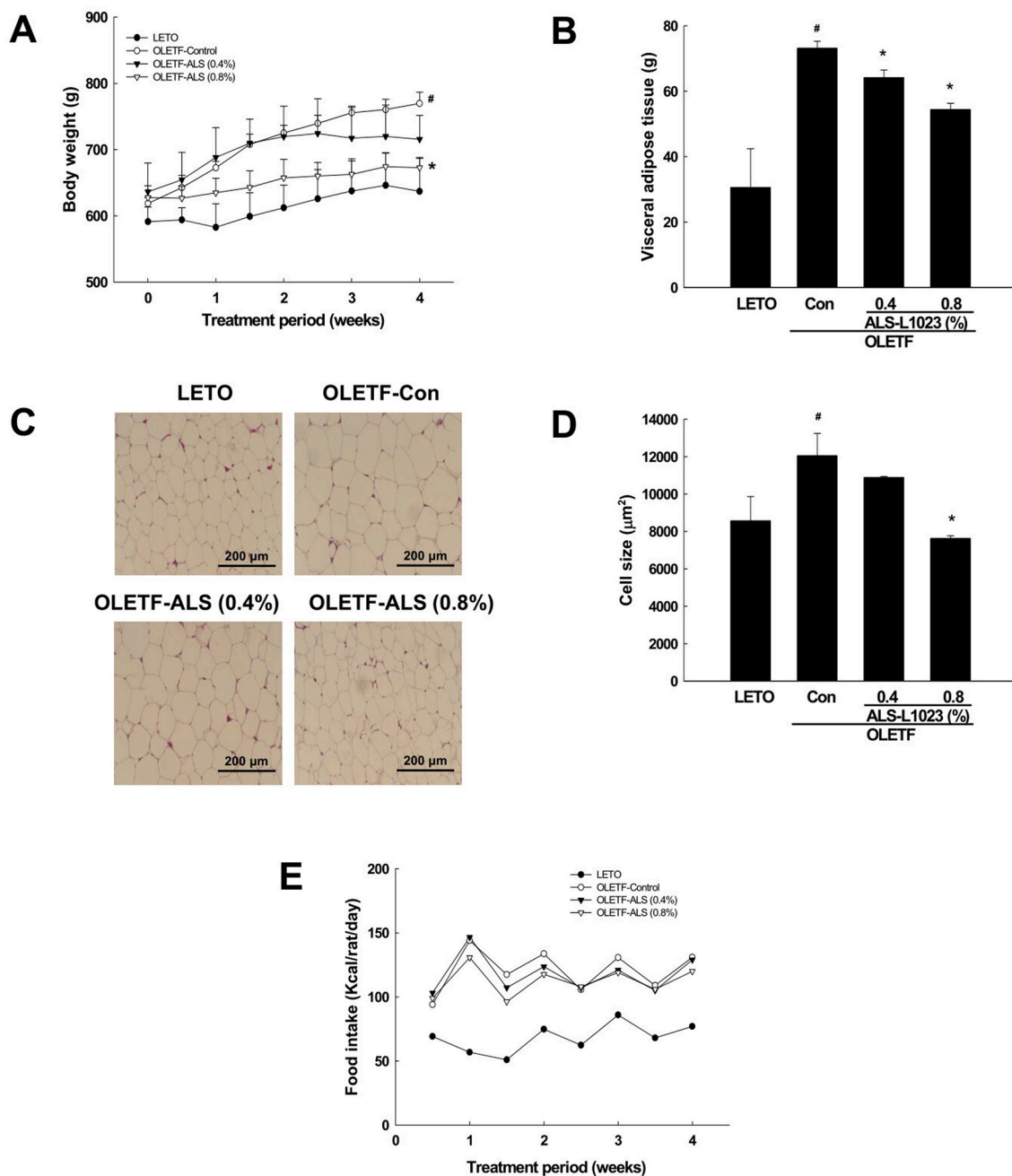


Fig. 1. Effects of ALS-L1023 on obesity, adipocyte hypertrophy, and appetite in diabetic OLETF rats. LETO and OLETF rats at 28 weeks of age were fed a Chow, an HFD or an HFD supplemented with 0.4 or 0.8% ALS-L1023 for 4 weeks. (A) Body weights, (B) visceral adipose tissue mass, (C) histological analysis of visceral adipose tissue, (D) visceral adipocyte size, and (E) food consumption profiles. # $p < 0.05$ compared to LETO. * $p < 0.05$ compared to OLETF-Con.

decreased the visceral (i.e., epididymal, mesenteric, and retroperitoneal) adipose tissue mass in obese rats (Fig. 1B). Supplementation with 0.8% of ALS-L1023 reduced the average size of the visceral adipocytes to a level similar to that of the lean LETO rats (Fig. 1C and D) without affecting food intake (Fig. 1E). In pair-feeding experiments, OLETF-ALS rats weighed less than pair-fed rats with a similar caloric intake (data not shown). No ALS-L1023 dose used in the present study showed toxic effects. These results suggest that ALS-L1023 regulates obesity and adipocyte hypertrophy in obese OLETF rats.

3.2. ALS-L1023 improves lipid and glucose metabolism

Consistent with weight loss, treatment of OLETF rats with 0.8% of ALS-L1023 resulted in reduced serum triglyceride and free fatty acid concentrations compared with the obese OLETF-Con rats (Fig. 2A and B). Obese OLETF rats showed a 239% increase in the fasting glucose levels and an 85% decrease in the serum insulin levels compared with lean LETO rats (Fig. 2C and D). However, treatment with 0.8% of ALS-L1023 resulted in a 45% decrease in the fasting glucose levels and a 242% increase in serum insulin levels compared with the obese OLETF-Con rats. These results indicate that ALS-L1023 markedly inhibits hyperglycemia, hypoinsulinemia, and dyslipidemia in diabetic rats.

3.3. ALS-L1023 treatment restores pancreatic islet β -cell mass

The H&E-stained pancreas sections from OLETF-Con rats showed markedly disorganized morphology without islet contours. By contrast, rats that received supplementation of 0.4% of ALS-L1023 showed slightly improved islet morphology, while those that received 0.8% of ALS-L1023 resulted in an islet morphology of relatively round

appearance (Fig. 3A). To examine the effects of ALS-L1023 on the islet β -cell mass, pancreas sections were stained with an antibody against insulin. Pancreas tissue from obese OLETF-Con rats showed islet β -cell destruction and a lower number of β -cells than in the LETO rats, as demonstrated by the decreased insulin-positive β -cells (Fig. 3B and C). However, areas of insulin-positive β -cell were increased in OLETF rats treated with 0.8% of ALS-L1023 compared with the untreated OLETF-Con rats. These results suggest that ALS-L1023 reduces β -cell death in a diabetic state.

3.4. ALS-L1023 reduces hepatic lipid accumulation

To determine whether ALS-L1023 regulates hepatic steatosis, we examined the Oil red O-stained intrahepatic triglyceride content following ALS-L1023 treatment. Although lipid accumulation was evident in the liver of OLETF-Con rats, treatment with ALS-L1023 reduced hepatic triglyceride levels (Fig. 4A and B). To evaluate whether the ALS-L1023-induced inhibition of hepatic lipid accumulation was caused by alterations in the hepatic expression of fatty acid oxidative enzymes, we measured the mRNA levels of AMP-activated protein kinase α 2 (AMPK α 2), acyl-CoA oxidase (ACOX), and very long-chain acyl-CoA dehydrogenase (VLCAD). They are enzymes that promote fatty acid β -oxidation in mitochondria and peroxisomes. The mRNA expression of all these genes was much higher in the ALS-L1023-treated OLETF rats than in the untreated OLETF-Con rats (Fig. 4C). The peroxisome proliferator-activated receptor α (PPAR α) mRNA levels remained unchanged by ALS-L1023 treatment. These results suggest that ALS-L1023 inhibits hepatic lipid accumulation by stimulating the expression of fatty acid oxidation enzyme genes.

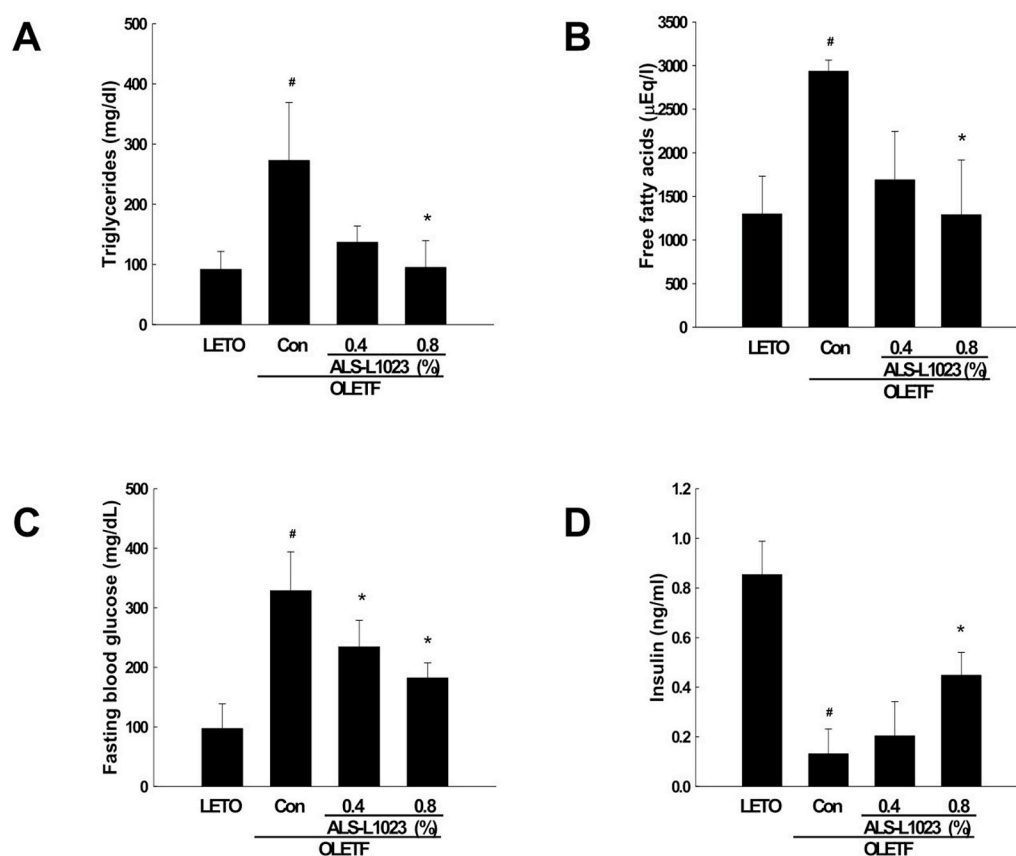


Fig. 2. Effects of ALS-L1023 on circulating levels of lipids, glucose, and insulin in diabetic OLETF rats. LETO and OLETF rats at 28 weeks of age were fed a Chow, an HFD or an HFD supplemented with 0.4 or 0.8% ALS-L1023 for 4 weeks. (A) Serum levels of triglycerides and (B) free fatty acids. (C) Fasting blood glucose and (D) serum insulin levels. [#]*p* < 0.05 compared to LETO. ^{*}*p* < 0.05 compared to OLETF-Con.

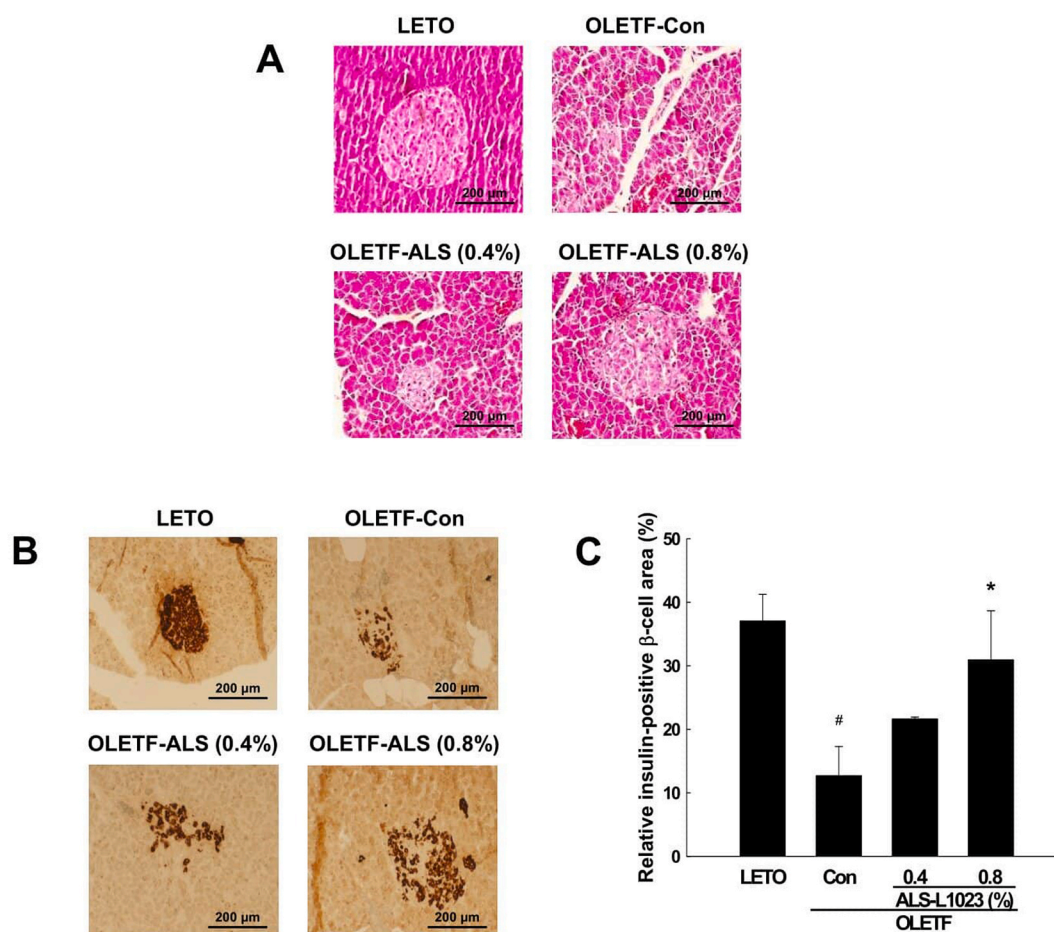


Fig. 3. Effects of ALS-L1023 on pancreatic morphology and insulin-secreting β -cell area in diabetic OLETF rats. LETO and OLETF rats at 28 weeks of age were fed a Chow, an HFD or an HFD supplemented with 0.4 or 0.8% ALS-L1023 for 4 weeks. (A) H&E-stained pancreatic sections (original magnification, $\times 100$). (B) Pancreas sections stained with an anti-insulin antibody (original magnification, $\times 100$). (C) Relative insulin-positive β -cell area. # $p < 0.05$ compared to LETO. * $p < 0.05$ compared to OLETF-Con.

3.5. ALS-L1023 decreases skeletal muscle lipid accumulation

To determine whether ALS-L1023 regulates skeletal muscle lipid accumulation, we measured intramyocellular lipid levels following ALS-L1023 treatment via Oil red O staining. OLETF-Con rats showed higher lipid accumulation than the LETO rats, as demonstrated by increased Oil red O staining. Conversely, supplementation with ALS-L1023 decreased the triglyceride content (Fig. 5A and B). To evaluate whether the ALS-L1023-induced inhibition of lipid accumulation in skeletal muscle was associated with alterations in skeletal muscle expression of fatty acid oxidation genes, we measured the mRNA levels of AMPK α 2 and medium-chain acyl-CoA dehydrogenase (MCAD). The expression levels of AMPK α 2 and MCAD were higher in the ALS-L1023-treated rats than in the untreated obese rats (Fig. 5C). The PPAR α mRNA levels were not different between untreated and treated OLETF rats. These results suggest that ALS-L1023 reduces fat deposition in skeletal muscle by increasing the skeletal muscle expression of genes responsible for fatty acid oxidation.

3.6. ALS-L1023 inhibits pancreatic inflammation

To determine whether ALS-L1023 modulates pancreatic inflammation under obese conditions, we examined immune cell infiltration and inflammatory factor expression in the pancreas of obese OLETF rats following ALS-L1023 treatment. Pancreas sections were stained with an antibody against CD68, which is expressed on macrophages and monocytes. The sections were also stained with toluidine blue to detect

mast cells. Infiltration of macrophages (Fig. 6A and B) and mast cells (Fig. 6C and D) into pancreatic islets was evident in the OLETF-Con rats compared with that detected in the LETO rats. However, there was a decrease in macrophages and mast cells in OLETF-ALS rats compared with that detected in the OLETF-Con rats. Pancreatic islets in T2D showed an elevated production of inflammatory factors, including tumor necrosis factor α (TNF α), CD68, and interleukin 6 (IL-6) (Ehse et al., 2007). Gene expression analysis showed that treatment with ALS-L1023 decreased the mRNA levels of CD68 and IL-6 in diabetic rats (Fig. 6E). These results indicate that ALS-L1023 may mitigate pancreatic inflammation by decreasing the expression of inflammatory factors.

4. Discussion

The present results demonstrate that the lemon balm extract ALS-L1023 reduced weight gain and glucose levels, in addition to reversing the loss of insulin-secreting islet β -cells. This phenomenon resulted partly through the reduction of ectopic lipid levels and pancreatic inflammation in diabetic OLETF rats.

The OLETF rat is a commonly used animal model of obesity and T2D. These rats have a satiety deficiency that leads to overall hyperphagia and obesity (Moran et al., 1998; Hajnal et al., 2005). Supplementation with ALS-L1023 in OLETF rats resulted in the reduction of body weight, visceral fat weight, and visceral adipocyte size without altering food intake. The present data are supported by other reports from nutritionally obese C57BL/6 J mice treated with ALS-L1023 (Park et al., 2015; Woo et al., 2016; Kim et al., 2017). Thus, supplementation with

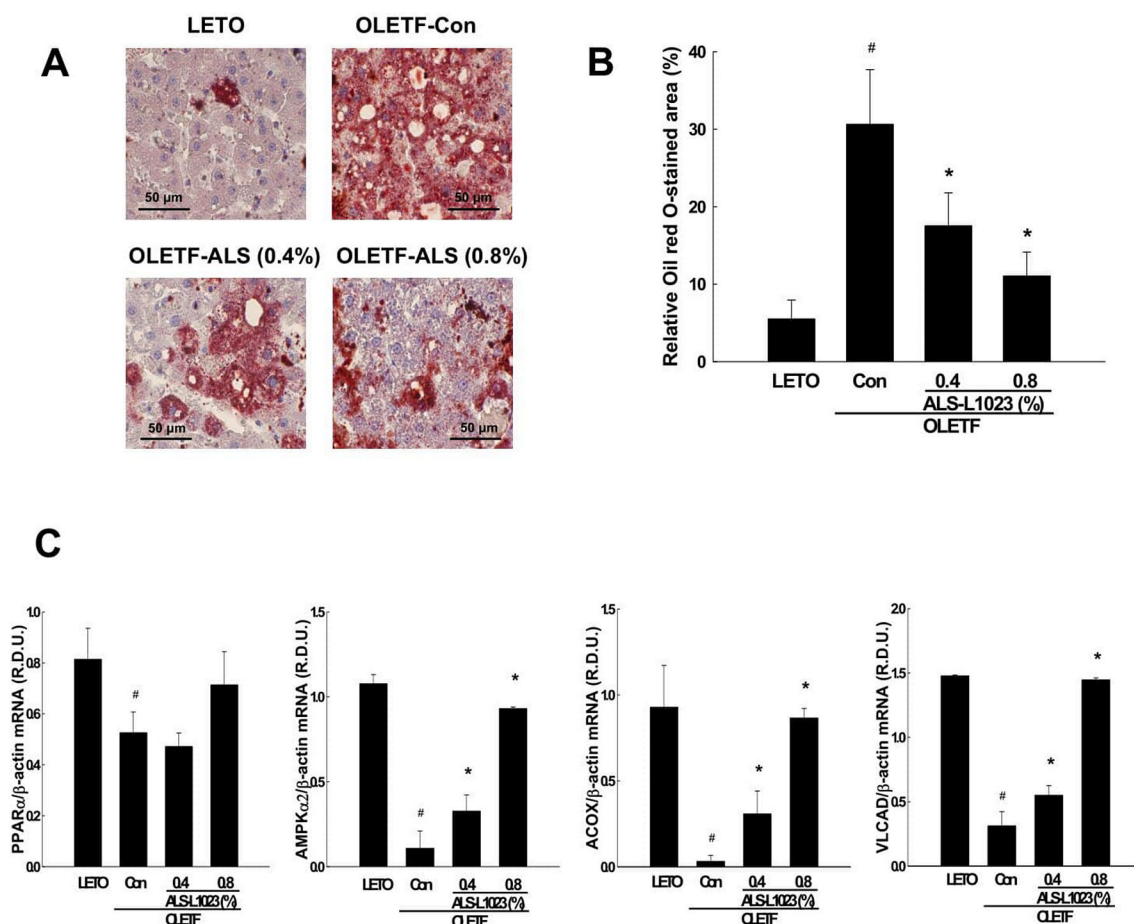


Fig. 4. Effects of ALS-L1023 on hepatic lipid accumulation in diabetic OLETF rats. LETO and OLETF rats at 28 weeks of age were fed a Chow, an HFD or an HFD supplemented with 0.4 or 0.8% ALS-L1023 for 4 weeks. (A) Oil red O-stained liver sections (original magnification $\times 400$). (B) Relative Oil red O-stained area in liver tissues. (C) The mRNA expression of genes responsible for fatty acid β -oxidation in liver tissues. # $p < 0.05$ compared to LETO. * $p < 0.05$ compared to OLETF-Con. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

ALS-L1023 inhibits visceral obesity and adipocyte hypertrophy without affecting appetite in both nutritionally and genetically obese animals. By contrast, 6-week treatments with the *Melissa officinalis* extract had no effect on body weight gain in C57BL/6 mice fed an HFD for an 18-week period (Weidner et al., 2014). The discrepancy in the results may be due to differences in the extraction methods applied to the leaves of *Melissa officinalis*, treatment periods and doses of *Melissa officinalis*, and the animal species used for analysis.

Excess energy intake causes an elevation in the concentrations of circulating triglycerides and free fatty acids. The resulting circulating free fatty acids and triglycerides are responsible for the hypertrophy of adipose cells (Bourgeois et al., 1983; Chaput et al., 2000). Similarly, HFD-fed OLETF rats had severe hypertriglyceridemia and an elevation in circulating free fatty acids. However, administration of ALS-L1023 normalized the free fatty acids and triglyceride levels comparable to those of the lean animals. Similarly, the crude extract and individual components of *Melissa officinalis* also decreased plasma triglycerides and free fatty acids in HFD-fed obese mice and in the Dahl salt-sensitive rat model of genetic hypertension (Somova et al., 2003; Liao et al., 2013; Weidner et al., 2014). These results show that ALS-L1023 exhibits hypolipidemic effects in obese animals, which suggests a therapeutic role for *Melissa officinalis* in dyslipidemia and cardiovascular disease.

Nearly 100% of the OLETF rats have hyperglycemia and hyperinsulinemia as a result of islet cell hyperplasia and peripheral insulin resistance; therefore, OLETF rats spontaneously develop diabetic syndrome at 20 weeks of age (Kawano et al., 1992; Ishida et al., 1995). With continued aging, OLETF rats develop hypoinsulinemia resulting from

the deterioration of islet β -cells (Yabuki et al., 2006). In this study, the 28-week-old OLETF rats were fed an HFD for 4 weeks and showed a marked islet β -cell impairment (Seok et al., 2012). In particular, the serum insulin levels and the area of β -cells were surprisingly low in our study compared with those reported in other studies using OLETF rats aged 28–30 weeks. This may be due to the 4-week period of feeding the OLETF rats an HFD to exacerbate T2D. Compared with the LETO rats, the HFD-fed OLETF rats showed a significant increase in fasting blood glucose levels, a decrease in serum insulin levels, significant hyperglycemia, and marked hypoinsulinemia. However, the fasting glucose levels were decreased and the serum insulin levels were increased in OLETF rats treated with 0.8% of ALS-L1023 compared with those in the OLETF-Con rats. The hypoglycemic effects of *Melissa officinalis* were also observed in HFD-fed obese mice and in streptozotocin-induced diabetic rats (Weidner et al., 2014; Hasanein and Riahi, 2015), whereas an increase in insulin secretion was observed in HFD-fed streptozotocin-induced diabetic mice (Jang et al., 2009). These results suggest that ALS-L1023 exerts its potent antidiabetic effects by normalizing the glucose and insulin levels.

Obese OLETF rats have marked islet atrophy and a substantially decreased β -cell mass (Koh et al., 2003). Islets from HFD-fed OLETF-Con rats were severely damaged and had irregular contours that were not easily delineated. Insulin-positive cells were dispersed and extremely low in number. By contrast, the islet morphology was relatively round in the ALS-L1023-treated OLETF rats. In treated rats, treatment with 0.8% of ALS-L1023 profoundly increased the β -cell area, a finding that is consistent with the elevated circulating insulin levels. Previous studies

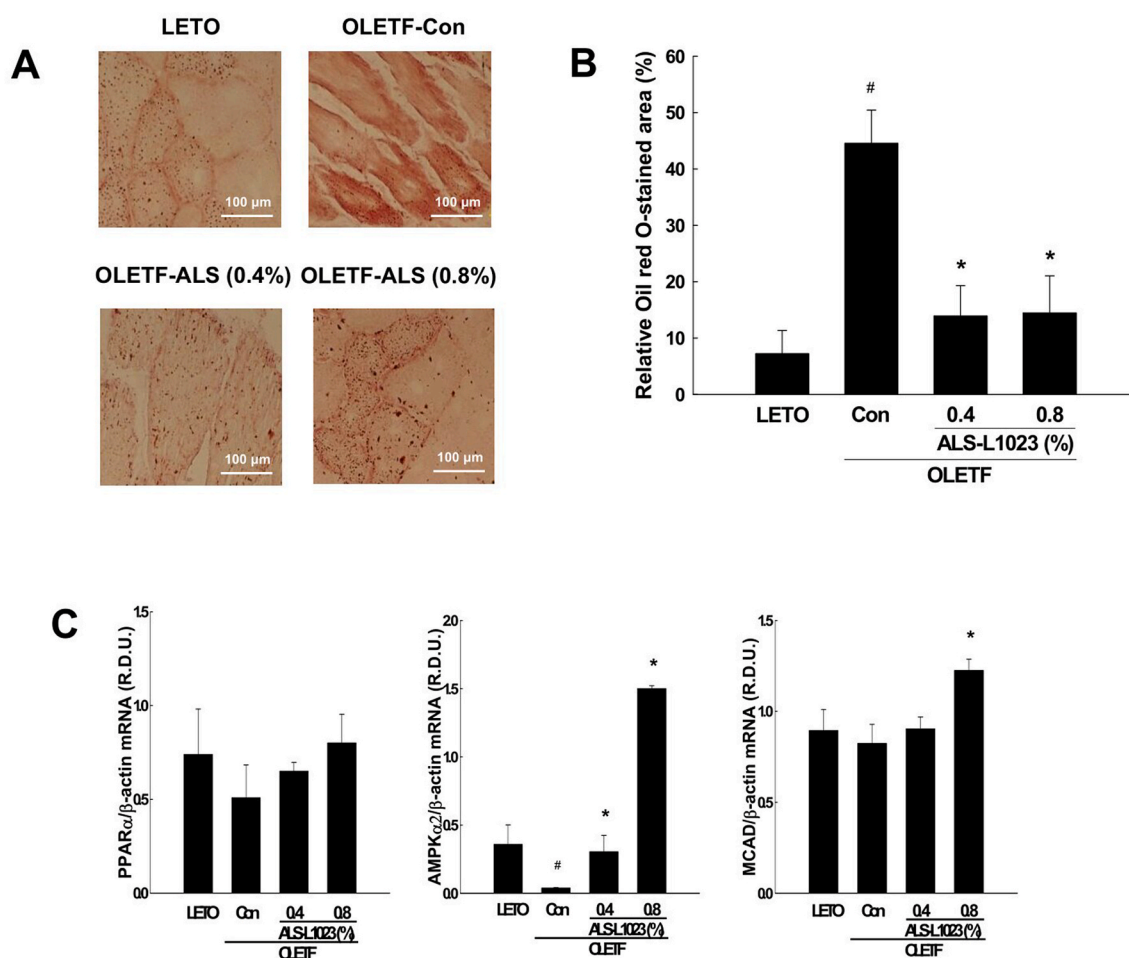


Fig. 5. Effects of ALS-L1023 on skeletal muscle lipid accumulation in diabetic OLETF rats. LETO and OLETF rats at 28 weeks of age were fed a Chow, an HFD or an HFD supplemented with 0.4 or 0.8% ALS-L1023 for 4 weeks. (A) Oil red O-stained sections of skeletal muscle (original magnification $\times 100$). (B) Relative Oil red O-stained area in skeletal muscle tissues. (C) Expression of genes responsible for fatty acid β -oxidation in skeletal muscle tissues. [#] $p < 0.05$ compared to LETO. ^{*} $p < 0.05$ compared to OLETF-Con. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

have reported that a triterpenoid constituent of *Melissa officinalis* increased the survival of pancreatic β -cells (Castellano et al., 2013). These results suggest that supplementation with ALS-L1023 may suppress the development of T2D and mitigate diabetic complications by reducing β -cell destruction.

Triglycerides can be stored in non-adipose tissues, such as the muscle, liver, and heart. Lipid droplets accumulate in the cytoplasm, and an excess lipid accumulation may result in cell dysfunction or cell death (van Herpen and Schrauwen-Hinderling, 2008). The accumulation of ectopic lipids in the liver and skeletal muscle is thought to cause insulin resistance and T2D (Perry et al., 2014; Broskey et al., 2018). An increased intramyocellular lipid content is associated with an impaired insulin-stimulated glucose uptake in obese rats, as well as in obese and healthy humans (Storlien et al., 1991; Pan et al., 1997; Broskey et al., 2018). Intrahepatocellular lipids are characterized by several features of insulin resistance in normal-weight and moderately overweight subjects (Seppälä-Lindroos et al., 2002). These factors are independent of body mass index, intraabdominal lipids, and overall obesity, although abdominal obesity is a marker of ectopic fat accumulation (Smith, 2015). Thus, we examined whether ALS-L1023 reduces excess lipid accumulation in the skeletal muscle and liver of OLETF rats. Our histological analyses revealed that OLETF-Con rats had marked lipid accumulation in the liver and skeletal muscle compared with the LETO rats. However, these lipid droplets were markedly decreased in both the organs of OLETF-ALS rats compared with the untreated OLETF-Con rats. These data suggest that ALS-L1023 inhibits excess lipid accumulation in

the liver and skeletal muscle, which can alleviate insulin resistance and T2D. In addition, ALS-L1023-induced reductions in fat deposition in the liver and skeletal muscle may contribute to an increased fat migration from adipose tissue to non-adipose tissues, resulting in the inhibition of weight gain.

Similarly, dysregulation of fatty acid oxidation may be a predictor of weight gain and T2D progression (Unger, 1995; McGarry, 2002). Liver and skeletal muscle tissues are major action sites of fatty acid oxidation. The results from previous studies demonstrated that the PPAR α agonist fenofibrate induces the mRNA expression of mitochondrial and peroxisomal enzymes in the liver and skeletal muscle of HFD-fed obese mice (Jeong et al., 2004; Lee and Yoon, 2013). The ligand-bound PPAR α heterodimerizes with the retinoid X receptor and then modulates the expression of target genes that contain a PPAR response element in their promoter regions. The target genes of PPAR α include those related to the hydrolysis of plasma triglycerides, fatty acid uptake and binding, and fatty acid oxidation (Yoon, 2009). Thus, activation of PPAR α target genes promotes fatty acid oxidation. In the present study, we investigated whether ALS-L1023 regulated the mRNA expression of PPAR α target genes such as AMPK α 2, MCAD, VLCAD, and ACOX. Consistent with the reduced lipid content in the liver and skeletal muscle, ALS-L1023 increased the expression of fatty acid-oxidizing enzymes but not PPAR α in both organs of the OLETF control rats. Therefore, ALS-L1023 may reduce ectopic lipid accumulation by increasing the expression of PPAR α target enzymes involved in fatty acid β -oxidation through the stimulation of PPAR α transactivation.

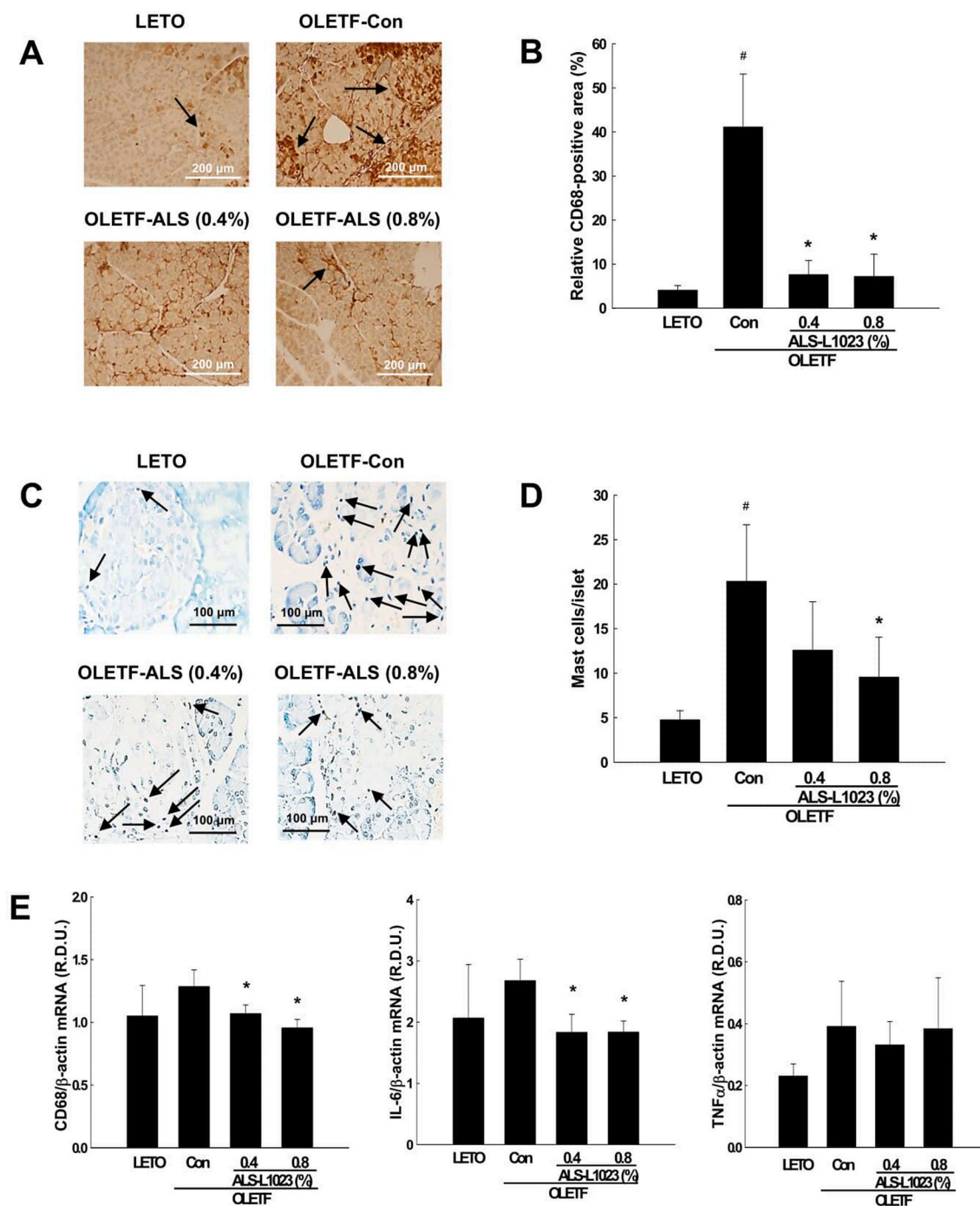


Fig. 6. Effects of ALS-L1023 on pancreatic inflammation in diabetic OLETF rats. LETO and OLETF rats at 28 weeks of age were fed a Chow, an HFD or an HFD supplemented with 0.4 or 0.8% ALS-L1023 for 4 weeks. (A) Immunohistochemical detection of CD68-positive macrophages in pancreatic tissues (original magnification $\times 100$). Arrows indicate macrophages. (B) Relative CD68-positive area. (C) Toluidine blue-stained sections of pancreatic tissues (original magnification $\times 400$). Arrows indicate mast cells. (D) The number of mast cells per islet. (E) The mRNA expression of inflammatory genes in pancreatic tissues. [#] $p < 0.05$ compared to LETO. ^{*} $p < 0.05$ compared to OLETF-Con. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

To further determine the mechanism by which ALS-L1023 inhibits hyperglycemia and alleviates β -cell loss, we also investigated pancreatic islet inflammation in ALS-L1023-treated OLETF rats. Inflammatory changes, including the accumulation of macrophages, have been documented in T2D islets (Eguchi and Manabe, 2013) and contribute to β -cell impairment (Donath et al., 2005, 2009). Increased numbers of islet-associated, CD68-positive macrophages were observed in T2D patients and animal models of T2D, such as GK rats and *db/db* mice (Ehse et al., 2007). Similar to previous reports, CD68-positive macrophages

and toluidine blue-stained mast cells were present at significantly higher number in the OLETF-Con rats than in the LETO rats. However, the numbers of macrophages and mast cells were markedly decreased in the OLETF-ALS rats compared with those in the OLETF-Con rats. These results are in accordance with the restoration of islet β -cell mass following ALS-L1023 treatment. Islets from T2D patients secrete inflammatory cytokines, such as IL-6 and IL-8 (Ehse et al., 2007), and the IL-1 receptor-antagonist treatment improves β -cell function, hyperglycemia, and inflammatory marker expression (Larsen et al., 2007). The

OETF-Con rats had an increased mRNA expression of IL-6, TNF α , and CD68 compared with the LETO rats. Conversely, the mRNA levels of IL-6 and CD68 were reduced in OETF-ALS rats compared with those in the untreated OETF-Con rats. These results suggest that ALS-L1023 reduces pancreatic islet inflammation, likely contributing to the decrease in hyperglycemia and the increase in β -cell survival.

Phytochemical studies on *Melissa officinalis* have revealed that it comprises several phytochemicals, such as essential oil, triterpenes, phenolic acids, flavonoids, and tannins (Shakeri et al., 2016). The pharmacological and medicinal activities of *Melissa officinalis* are mainly due to essential oil (i.e., geranial, neral, citronellal, and geraniol), phenolic acids (i.e., caffeic acid and rosmarinic acid), and triterpenes (i.e., ursolic acid and oleanolic acid) (Schnitzler et al., 2008; Awad et al., 2009). The active components of *Melissa officinalis* exhibit antianxiety, antidepressant, anti-inflammatory and antioxidant effects. These active components have also demonstrated hypoglycemic and hypolipidemic effects (Jang et al., 2009; Castellano et al., 2013; Liao et al., 2013; Weidner et al., 2014; Alam et al., 2016; Seyedan et al., 2017), suggesting that *Melissa officinalis* may inhibit obesity, dyslipidemia, and diabetes. This process may be attributed to its active components, such as essential oil, rosmarinic acid, and caffeic acid. Therefore, further studies are necessary to identify and characterize the active components of ALS-L1023 and to determine their respective effects on obesity and diabetes.

5. Conclusion

In conclusion, our results suggest that the herbal extract ALS-L1023, isolated from the leaves of *Melissa officinalis*, may not only reduce weight gain and blood glucose levels but also inhibit islet β -cell destruction in obese diabetic OETF rats, and contributes to the amelioration of obesity and T2D development. These effects may be mediated by regulating the expression of genes involved in fatty acid β -oxidation in the liver and skeletal muscle, in addition to its effects on pancreatic inflammation. Thus, ALS-L1023 may be a potential therapeutic approach to treat human obesity and T2D.

Declaration of competing interest

None declared.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2020.113360>.

Author contributions

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